A Sequential Stochastic Approach to Rationalizing Hedging Decisions in Agricultural Markets

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Abstract

Farmers have historically exhibited a tremendous amount of heterogeneity in their hedging patterns. Previous research has mostly modeled farmer behavior by employing the mean-variance hedge ratio in a 2-period investment setting. In this paper, we use expected utility theory to rationalize hedging decisions made by farmers in a dynamic setting, allowing for subjective beliefs about market conditions. We enable our representative farmer to make hedging decisions on a weekly basis to account for incremental changes in market beliefs. We find that subjective prior beliefs and crop insurance coverage play a significant role in hedging decisions. Other factors such as initial wealth play a small but non-negligible role in guiding hedging decisions. Our results indicate that only pessimistic farmers hedge in concordance with traditional financial wisdom. We also find that unbiased farmers, regardless of the degree of risk aversion, forgo hedging until late in the calendar year.

Keywords: Hedging; Risk; CRRA; Futures; Crop Insurance.

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1 Introduction

It is well known that corn futures exhibit strong seasonality (CME Group, 2022; Sørensen, 2002). Early in the growing season, when supply is low and uncertainty is high, prices are typically higher. Historically, as harvests commence and uncertainty resolves, corn futures prices begin to fall as they converge to the spot price at the end of the growing season. Thus, not all hedging decisions should be treated the same. Indeed, an 80% hedge in early June would, on average, be much more financially prudent than a 80% hedge in October when futures prices are expected to fall. Given the historical seasonality of corn futures prices, we contend that the timing of the hedge is just as imperative, if not more, than the total volume hedged at the end of the season. Figure 1 shows the average weekly corn futures price in dollars per bushel from 2004-2024 and illustrates the importance of analyzing hedging decisions in a sequential manner. Thus, we generalize standard hedging models to allow framers to hedge each week over a six month period. Our time frame consists of 27 weeks. The first period corresponds to the first week of June, when planting is typically completed across the United States. Our terminal date is the last week of November, when the December corn contract is near expiration.

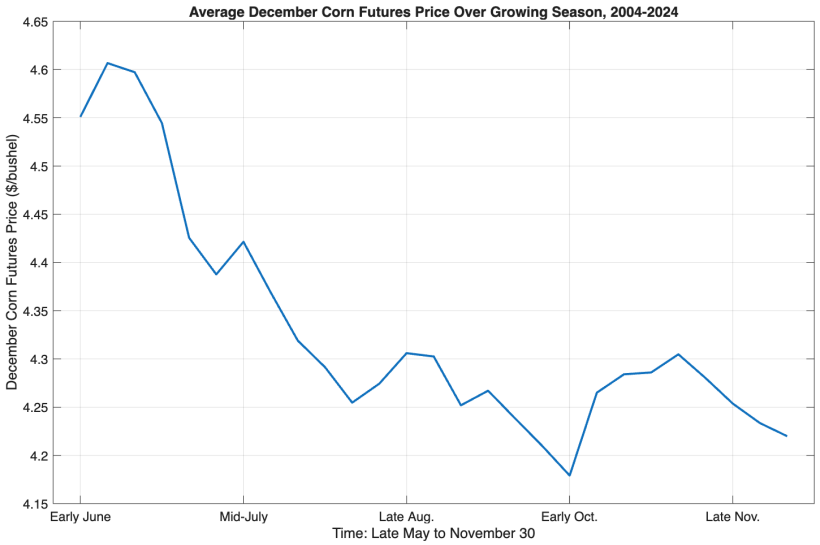


Figure 1: Average December Corn Futures Price Over Growing Season, 2004-2024

Note: Data from Barchart

Additionally, hedging behavior in farmers has been seen as low compared to what conventional financial wisdom would suggest (Prager et al., 2025; Tomek and Peterson, 2001). Data from Prager et al. (2020) show that, in 2016, just 9.9% of corn and soybean producing farms in the United States hedged their corn product. The farmers that hedged their corn sold on average 41.1% of their product on the futures market. Only 6% of farmers utilizing the futures market hedged the entirety of their crop. Moreover, larger corn farms are much more likely to trade in options.

Past research has advanced a number of explanations for low levels of observed hedging. Dorfman and Karali (2010) find that habit, defined as the connection between current and lagged hedging decisions, plays a significant role in hedging decisions. High basis risk and greater cash constraints are also connected with lower levels of hedging (Prager et al., 2020). Farmers with more certainty about prices, lower levels of risk aversion and diversified investment portfolios tend to hedge less (Pannell et al., 2008). Larger farms tend to trade more frequently in futures and options, along with younger farmers and farmers with college educations (MacDonald, 2020). Pennings and Leuthold (2000) found that risk attitude played a significant role in determining the extent to which farmers hedged in the futures market.

Moreover, previous literature on hedging decisions focuses significantly on the mean-variance hedge ratio framework, namely:

$$h^* = \frac{Cov(\Delta S, \Delta F)}{Var(\Delta F)} \tag{1}$$

where h^* is the optimal hedge ratio, ΔS = the change in spot price and ΔF = the change in futures prices. This methodology was proposed by Johnson (1960), positing that the solution is the optimal hedge ratio. The popularity of this model accelerated as it garnered empirical validity (Ederington, 1979).

However, this deterministic formula fails to account for subjective measures such as risk preference and prior beliefs regarding the direction and magnitude of future price changes

throughout the season. Subjective probability can be dated back to Ramsey's *Truth and Probability* (Ramsey, 1931), where he proposed that probability is not objective in the frequentist sense, but expressed as a subjective degree of belief contingent upon prior beliefs. In *General Theory* (Keynes, 1936), Keynes argued that the "animal spirits", or one's subjective economic outlook, played a heavy role in one's investment decisions. Many previous hedging models fail to account for said subjectivity. Implementing subjective parameters into our expected utility model will be key in rationalizing these investment decisions.

We also wish to characterize a time path of hedging, along with a rationalization of the heterogeneous hedging behavior under subjective expectations. Specifically, we attempt to further rationalize hedging behavior in a sequential dynamic stochastic setting that accounts for subjective beliefs about the projected futures prices throughout the season. We endow our representative farmer with a von Neumann-Morgenstern constant relative risk aversion (CRRA) expected utility function that is optimized weekly over 27 weeks, which is taken to be the hedging window of the December corn contract.

Many previous papers model structural hedging decisions in a two-period time setting (Lapan and Moschini, 1994; Pannell et al., 2008). Hedging was later introduced through a dynamic sequential framework, allowing for the characterization of a time path towards an optimal hedge (Anderson and Danthine, 1983). Martinez and Zering (1992) formulated a dynamic hedging model and used econometric techniques to forecast futures prices and inform incremental hedging decisions and found that the gains from dynamic hedging were positive but economically negligible. Myers and Hanson (1996) extended the analysis of Anderson and Danthine (1983) by introducing a discrete dynamic hedging problem that relaxes assumptions regarding the underlying utility function and price distributions. Our sequential dynamic approach introduces the assumption that farmers' hedging habits might change on a week by week basis as prices and their prior beliefs on the state of the corn market change. This relaxes the restrictive assumptions of the two period model and enables farmers to adjust hedging decisions as new information arrives throughout the growing season.

In our model, the farmer's utility is a function of a certain current futures price, uncertain futures price later on in the growing season, previous futures sales, the expectation of the spot price at the conclusion of the season, and initial wealth. The flexible nature of the CRRA utility function enables us to fluctuate the Arrow-Pratt coefficient of relative risk aversion to assess how risk aversion levels affect aggregate hedging decisions over a futures contract season. At t = 1, the farmer will optimize his utility over the current week and the remaining hedging season, where each week from 2 to 27 is denoted with a distribution to represent the farmer's beliefs about the future prices. At t = 2, the farmer's week 1 decision will be internalized as he plans from week 3 to 27. This process iterates until week 27 when the contract expires, or the farmer has hedged the entirety of their corn.

Additionally, we operate under a non-stochastic production function to isolate the effects of risk aversion and other parameters of interest. It is well known in agricultural economics that production functions are stochastic, given the nature of drought and pest infestations. However, introducing stochasticity into the production function results in lower hedge ratios, as shown by Lapan and Moschini (1994). Thus, we assume non-stochastic production to isolate the effects of price risk on expected utility functions under subjective expectations and to emphasize the fact that farmers may delay/forgo hedging for reasons outside of stochastic production.

For prior distributions, we will fluctuate both mean and standard deviation of our representative farmer's distribution of future expected corn prices to assess their impact on final hedging decisions. The prior mean represents a farmer's level of bearishness or bullishness – subtracting from the mean denotes bearishness, whereas adding to the mean represents bullishness. Similarly, standard deviation measures subjective uncertainty. We also allow for our farmers to purchase revenue insurance, as Coble et al. (2000) has shown that farmers hedge significantly less of their production when covered by insurance.

Corn futures are one of the most frequently traded agricultural commodities with approximately 1.75 billion bushels being traded daily CME Group (2026a). Moreover, corn

futures are an imperative tool for farmers to mitigate price risk. The proliferation of revenue insurance, introduced through the Federal Crop Insurance Program in 1996, has also served as an important tool for risk management. In 2024, $13.2 billion dollars were spent by the federal government on crop insurance program payouts Congressional Budget Office (2025). Thus, understanding how a farmer would rationally hedge with the availability of both of these risk mitigation tools, along with assessing the interaction between public and private insurance, is a worthwhile pursuit in agricultural economics.

To our knowledge, this is the first paper that rationalizes farmers' hedging decisions in a dynamic framework that allows for both insurance contracts and subjective parameters in the agent's expected utility function, emphasizing the importance of hedge timing in a sequential setting. Moreover, we isolate the effects of changes in moments of the agent's belief distribution and report their effects on both the timing and the optimal hedge ratio at the end of the season. We also allow our representative farmer to pair hedging with crop insurance to manage risk, and assess the degree of substitutability between insurance and hedging. The paper's key contribution is demonstrating that this more complete model of farmer behavior can easily rationalize the heterogeneous nature of observed hedging behavior, demonstrating multiple explanations of why rational farmers do not hedge more.

The paper proceeds as following. We next present the model that our representative farmer optimizes. We then explain how we operationalize the farmer's prior beliefs through distributions. Next, a comparative statics section is presented, where we characterize how the farmer will react as we fluctuate the model's parameters, *ceteris paribus*. We then explain how and why we chose the model parameters. A results section follows, where we summarize both the season ending hedge ratios, and analyze how the farmer chooses to allocate their chosen hedge ratios across our model's time span. We conclude the paper with a sensitivity analysis.

2 The Model

The representative farmer gains utility from initial wealth, futures sales at time t, previous future sales earlier in the season at week k, planned futures sales at week j, and planned sales on the local spot market in the final time period. Let f_t be the futures price at week t. Let $\tilde{f}_j$ be the predicted futures price at week j. The farmer has a prior distribution for each week in the season. Thus, at time period t, the farmer has a total of $T-t$ prior distributions. For our model, $t = 1$ is the last week of May, and $T = 27$ is the expiration of the December contract. Let $\tilde{P_T}$ be the predicted spot price at the end of the December contract. Let W_0 be the initial wealth endowed to the farmer. The farmer has the following variables: θ_t, a hedge ratio sold at week t, and a time-varying $(T-t) * 1$ vector of planned hedge ratios, $\tilde{\theta_j}$ to be sold at week j. The $\tilde{\theta_j}$ parameters are not locked into the farmer's final hedging decision, only θ_t is. However, $\tilde{\theta_j}$ helps the farmer plan and internalize future hedging decisions. The vector of θ_t and $\tilde{\theta_t}$ for $t = 1, 2, ..., T$ must lie on or inside the unit simplex, restricting the farmer to futures positions representing partial or full hedging but not speculating. Let ρ equal the Arrow-Pratt coefficient of relative risk aversion. We also allow the farmer to have subjective expectations about the trajectory of the futures prices. Let o_j be an optimism parameter. If the farmer is optimistic, then he adds 1 cent to the mean of the predicted futures price at each week j. In other words, if a farmer is optimistic in week 1, then the predicted futures price in week 2 will have one cent added to it, the futures price in week 3 will have two cents added to it, and so forth, culminating with $T-t$ cents being added to the predicted spot price. We also allow our farmer to be pessimistic, where we subtract 1 cent instead of add. Throughout the season, the farmer solves $T-t$ optimization problems, one for each week. The hedge ratio at the end of the season, $T = 27$ is defined as $\sum_{t=1}^{T} \theta_t$.

It is well known that a risk averse agent who faces only price risk will fully hedge, regardless of the magnitude of risk aversion (Holthausen, 1979). However, this result only holds if the stochastic price later on in the season has the exact same mean as the current future price. To guarantee an interior solution, we generate 10,000 draws from $\tilde{f}_j$ and $\tilde{P_T}$.

The finite nature of the distributions ensures that there is sufficient differentiation from the mean, thus driving an interior solution. This also underscores the importance of the subjective beliefs of our representative farmer; indeed, the farmer need not believe that the futures or spot price later on in the season will be exactly the same as the current futures price.

Our representative farmer is endowed with a CRRA von Neumann-Morgenstern utility function. The non-negativity requirements of the hedge ratios necessitate the usage of Karush-Kuhn Tucker conditions. Let λ, η_1, η_2 be Lagrange multipliers. The function is maximized by choosing weekly hedging positions according to: [1]

$$\begin{aligned}
&\max_{\theta_t\,\{\tilde{\theta}_j\}_{j=t+1}^{T}} \quad \mathbb{E}_t\left[\frac{1}{1-\rho}\left(W_0 + \underbrace{f_t\theta_t}_{\substack{\text{Current}\\ \text{futures revenue}}} + \underbrace{\sum_{k=1}^{t-1} f_k\theta_k}_{\substack{\text{Previous}\\ \text{revenue}}} + \underbrace{\sum_{j=t+1}^{T-1}(\tilde{f}_j + o_j)\tilde{\theta}_j}_{\substack{\text{Predicted}\\ \text{prices}}} + \underbrace{\left(1 - \sum_{k=1}^{t}\theta_k - \sum_{j=t+1}^{T-1}\tilde{\theta}_j\right)(\tilde{P}_T + o_{j,T})}_{\substack{\text{Spot}\\ \text{price revenue}}}\right)^{1-\rho}\right] \\
&\text{subject to} \quad \sum_{k=1}^{t}\theta_k + \sum_{j=t+1}^{T-1}\tilde{\theta}_j \leq 1, \\
&\text{Complementary slackness:} \quad \theta_t \geq 0, \quad \tilde{\theta}_j \geq 0, \quad \lambda \geq 0, \quad \eta_1 \geq 0, \\
&\qquad\qquad \eta_2 \geq 0, \quad \eta_1\theta_t = 0, \quad \eta_2\tilde{\theta}_j = 0, \quad \lambda\left(1 - \sum_{k=1}^{t}\theta_k - \sum_{j=t+1}^{T-1}\tilde{\theta}_j\right) = 0.
\end{aligned} \tag{2}$$

Let R be the revenue flow nested in the CRRA utility function. After establishing the Lagrangian, the first order conditions for maxima are:

$$\frac{\partial \mathcal{L}}{\partial \theta_t} = \mathbb{E}[\underbrace{(f_t - \tilde{P}_T)}_{\text{Basis Risk}} R^{-\rho}] + \lambda + \eta_1 = 0 \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial \tilde{\theta}_j} = \mathbb{E}[[(\tilde{f}_j - \tilde{P}_T)]\, R^{-\rho}] + \lambda + \eta_2 = 0 \tag{4}$$

The first order conditions impose basis risk upon the representative farmer, as illustrated

[1] Note that $E[U]$ decomposes to $E[log(U)]$ when $\rho = 1$, by L'Hôpital's Rule.

in equation (3). In other words, when deciding to hedge in the current time period, the farmer must face the tradeoff of selling their corn at a certain futures price or forgoing the sale and waiting to sell on the local spot market. We next elucidate the process of formulating our farmer's prior distributions.

3 Modeling Price Expectations

In our model, we use the December 2024 corn futures contract (ZCZ24) from the Chicago Board of Trade. While futures prices are often expressed in cents, we denote the futures price in dollars for computational purposes. We also normalize the number of bushels sold to 1 for computational simplicity. Figure 2 shows a time series of the weekly 2024 corn futures prices in dollars.

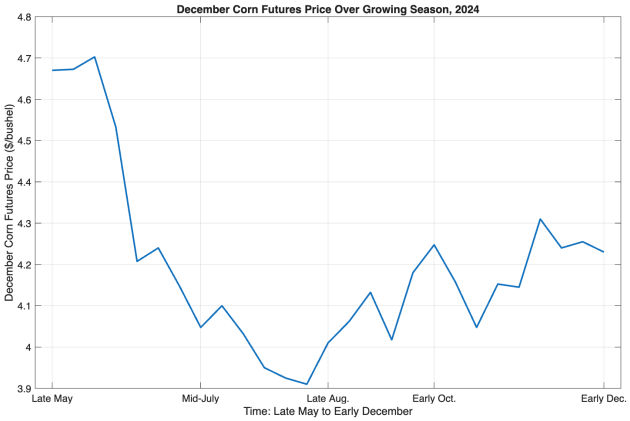


Figure 2: Time Series Plot of 2024 Futures Prices, December Contract

Note: Data from Barchart

For any given week t, we assume the mean of each forecasted distribution is the current futures price f_t.[2] For the forecasted prior standard deviation, we use the implied volatility for corn in 2024. Implied volatility has a long empirical history of being a strong predictor of futures volatility, especially in agricultural markets (Szakmary et al., 2003; Egelkraut et al., 2007; Wu et al., 2015). Adjusting the standard deviation for different horizons involves

multiplying the IV by the square root of time. Let σ_{IV} be the volatility factor, which was 0.19 in 2024 (Sherrick et al., 2024). Let t equal the current time period, and j equal a forward looking time horizon for weeks later on in the season. From CME Group (2026b), the standard deviation of the prior distribution for week j at time t is defined as:

$$\sigma_{t,j} = f_t * \sigma_{IV} * \sqrt{\frac{j}{52}} \tag{5}$$

This ensures that the farmer's prior variance starts out small in short time horizons, and expands as the time horizon extends towards T. Indeed, a rational farmer will have more certainty about next week's futures price compared to futures prices later on in the season.

Determining the distribution for the spot price of corn is a non-trivial task. Unlike futures prices, there is no single spot price for corn in the United States. Given that the spot price exists after the December futures contract has expired, we need to approximate it. Fama and French (1987) appeal to the Efficient Market Hypothesis and show that the current futures price is a strong, unbiased predictor of the spot price for any given commodity. Gardner (1976) extended this work to agricultural commodities as well. In light of these findings, we use the current futures price at time t as the mean for the spot price distribution. To determine the variance of the distribution of the spot price, we use equation (5) and set j equal to the number of remaining weeks until the contract expiration.

The standard deviation of the distribution at a later time period j, is given as $\sigma_{t,j}$ from equation (5) above. We also address the fact that not all farmers have the same level of subjective uncertainty. To do so, we introduce an uncertainty parameter, κ, that multiplies the standard deviation of the distribution to express different degrees of uncertainty. Lower levels of κ imply high levels of certainty about the trajectory of futures prices, whereas higher levels of κ represent a higher degree of uncertainty. For a given future time period j, the farmer's prior distribution assumes the form:

[2]The distributions of the expected futures prices need not be correlated with each other, as Samuelson (2015) proved that the next period's futures price differences are uncorrelated with the previous period's prices.

$$\tilde{f}_j \sim N(f_t, \kappa * \sigma_{t,j}) \tag{6}$$

$$\tilde{P}_T \sim N(f_t, \kappa * \sigma_{t,T}) \tag{7}$$

Lastly, we allow our farmer to be covered by revenue insurance. In 2024, the projected price for corn was \$4.66 (Paulson et al., 2024). Let I be the chosen amount of insurance coverage by the farmer. Under the Revenue Protection program, farmers are allowed to choose I starting from a value of 0.50, and increasing in increments of 0.05, with a maximum ceiling of 0.85. (USDA Risk Management Agency, 2026a). The Enhanced Coverage Option (ECO) enables farmers to choose coverage options up to 95% (USDA Risk Management Agency, 2026b). For computational purposes, we exploit the homothetic nature of the CRRA utility function and scale the output to 1 unit and interpret outcomes on a per-unit basis. Thus, for our insured farmer, we censor their prior distribution such that \$4.66*I is the minimum possible value. More specifically, our insured farmer's prior is:

$$\tilde{f}_j^* = \max\{\tilde{f}_j, 4.66 * I\}, \text{ where } \tilde{f}_j \sim N(f_t, \kappa * \sigma_{t,j}) \tag{8}$$

$$\tilde{P}_T^* = \max\{\tilde{P}_T, 4.66 * I\}, \text{ where } \tilde{P}_T \sim N(f_t, \kappa * \sigma_{t,T}) \tag{9}$$

Each $\tilde{f}_j$, $\tilde{P}_T$ distribution contains 10,000 draws of predicted prices. The way our farmer makes hedging decisions based on these prior distributions is heavily contingent upon structural parameters that dictate the shapes of these distributions. Thus, we proceed to comparative statics to assess their effects on these decisions.

4 Comparative Statics

Our farmer gains utility from current wealth, a sale in the futures market at price f_t, previous sales, futures prices that are subjectively projected later on in the season, $\tilde{f}_j$, initial wealth, and risk aversion. Because f_t is strictly positive, the farmers' utility is strictly increasing in

f_t. However, because $\tilde{f}_j$ is a distribution with moments that are contingent upon the presence of an insurance contract, the fluctuations of the moments are not entirely straightforward. In what proceeds, we will assess how changes in our structural parameters influence expected utility.

4.1 Variance - Uninsured Farmers

Because $\tilde{f}_j$ is a random variable, we can apply a second order Taylor expansion around the mean, $\mu = f_t$, to approximate $E[U(\tilde{f}_j)]$ as a summation of its moments. Specifically, if we let $\mu = f_t$,

$$\mathbb{E}[U(\tilde{f_j})] \approx \mathbb{E}[U(f_t) + U'(f_t)(\tilde{f}_j - f_t) + \frac{1}{2}U''(f_t)(\tilde{f}_j - f_t)^2] \tag{10}$$

$$\approx U(f_t) + U'(f_t)\underbrace{\mathbb{E}[\tilde{f}_j - f_t]}_{=0} + \frac{1}{2}U''(f_t)\underbrace{\mathbb{E}[\tilde{f}_j - f_t]^2}_{=\sigma^2_{\tilde{f}_j}} \tag{11}$$

From the second order Taylor expansion, variance can be approximated as

$$\frac{1}{2}\mathbb{E}[-\rho R^{-\rho-1}(f_t - \tilde{P}_T)^2]\sigma^2_{\tilde{f}_j} \tag{12}$$

Note that the mean of $\tilde{P}_T$ is also f_t. Thus, exploiting the distributive property of expected value functions enables variance to enter into the Taylor expansion twice - once from the variance of the expected futures price distribution, and once from the variance of the expected spot price distribution. As a result, variance multiplicatively enters twice into the Taylor approximation.

$$= \frac{1}{2}\mathbb{E}[-\rho R^{-\rho-1}]\ \underbrace{\mathbb{E}[(f_t - \tilde{P}_T)^2]}_{\sigma^2_{\tilde{P}_T}}\ \sigma^2_{\tilde{f}_j} \tag{13}$$

$$= \frac{1}{2}\mathbb{E}[-\rho R^{-\rho-1}]\sigma^2_{\tilde{P}_T}\sigma^2_{\tilde{f}_j} \tag{14}$$

If $\rho > 0$, then $U''(f_t) < 0$, as illustrated above. In other words, risk averse farmers view prior variance as a reduction in expected utility in the absence of insurance. Thus, uninsured risk averse farmers experience a decrease in utility with increasing variance. As variance increases, we can expect farmers to favor the current, risk free hedge θ_t over the $\tilde{\theta_j}$ that is multiplied to a distribution with a high degree of variance. As a result, more variance should result in a higher level of hedging for uninsured farmers.

4.2 Variance - Insured Farmers

Our insured prior distribution follows a censored normal. Let a be the revenue guarantee, namely, $a = I * \$4.66$. Let ${\tilde{f}_j}^* \sim N(f_t, \sigma^2)$ and $\tilde{f}_j = a$ if ${\tilde{f}_j}^* \leq a$ or else $\tilde{f}_j = {\tilde{f}_j}^*$. Let $\alpha = \frac{a - f_t}{\sigma}$ and $\phi(\alpha)$ be the standardized normal probability density function. Let $\Phi = \text{Prob}({\tilde{f}_j}^* \leq a)$ and $\lambda = \frac{\phi}{(1-\Phi)}$, the inverse Mills ratio. Following Greene (2018),

$$\mathbb{E}[\tilde{f}_j] = \Phi a + (1 - \Phi)(f_t + \sigma\lambda) \tag{15}$$

Because the expected value is expressed in terms of standard deviation and not variance, we substitute $\sigma = \sqrt{\sigma^2}$. Taking the partial derivative of (15) with respect to variance yields:

$$\frac{\partial \mathbb{E}[\tilde{f}_j]}{\partial \sigma^2} = \frac{\phi(\alpha)}{2\sigma} > 0 \tag{16}$$

Because $\phi(\alpha)$ is a probability density function, and σ is non-negative by definition, their quotient is non-negative. Thus, we can expect our insured farmer to hedge less with higher variance, as draws that fall below the price floor will be censored, and draws on the right tail will be uncensored. In other words, an increase in uncertainty increases the mean of the expected futures prices later on in the season. As these averages increase, the current futures price looks less attractive, and hedging is thus disincentivized. A more detailed proof

of (16)'s derivation can be found in the appendix.

4.3 Risk Aversion

The relationship between a risk averse agent's preferences over a gamble and a risk-free asset is best exemplified through Jensen's Inequality, namely:

$$\mathbb{E}[U(\tilde{f}_j)] < U[\mathbb{E}(\tilde{f}_j)]$$

Because $\mathbb{E}[\tilde{f}_j] = f_t$, this simplifies to:

$$\mathbb{E}[U(\tilde{f}_j)] < U[f_t]$$

where the inequality sign is a result of the risk averse nature of our representative farmer. Jensen's Inequality is a defining property of a concave function, and it expresses the notion that a risk averse individual prefers a certain outcome to an outcome that is random. Thus, with higher levels of risk aversion, we should expect our representative farmer to favor the certain futures price over the distribution of uncertain prices later in the sequence.

4.4 Initial Wealth

The notion of the CRRA utility function exhibiting Decreasing Absolute Risk Aversion (DARA) is a well-known result in microeconomic theory. Let R be the revenue stream in (6) above and $A(R)$ be the Arrow-Pratt coefficient of absolute risk aversion, defined as $A(R) = -\frac{u''(R)}{u'(R)}$. For CRRA utility, $A(R) = \frac{\rho}{R}$, and $\frac{\partial A(R)}{\partial R} < 0$, thus satisfying DARA. As a result, we should expect farmers to hedge less when endowed with higher levels of initial wealth.

The comparative statics above illustrate how our representative farmer will theoretically react to fluctuations in the model parameters. For variance, the comparative static sign is contingent upon the presence of an insurance contract. For uninsured farmers, increasing

variance is expected to lower the utility of planned future sales, thus resulting in higher hedging. For insured farmers, increasing variance does not result in more negative draws because of the censoring of the left tail of the distribution, but the more favorable positive draws are left unperturbed. This shifts the mean of the planned future sales to the right, resulting in lower hedging. Hedging will increase along with risk aversion and decrease with increases in wealth, as dictated by standard microeconomic theory. Due to the lack of a closed form solution for the CRRA utility function under normally distributed priors, simulations are warranted to derive optimal hedge ratios. We proceed to detail the process of solving these optimization problems.

5 Solving the Optimization Model

We solve the constrained optimization problem in equation (2) in MATLAB using the fmincon package with the parameters listed in Table 1.[2] These parameters yield 900 distinct combinations. For each of the 900 combinations, 1000 simulations across different distributional realizations are run and optimized. The average season ending hedge ratio is calculated across all 1000 simulations for each parameter combination.

Table 1 lists the parameters we employ in our optimization model. Initial wealth is calculated using USDA farm statistics. We choose initial wealth levels of \$25 and \$50 per bushel for our representative farmers. We assume that initial wealth is comprised of land valuation, machinery investment and off-farm assets, and we utilize USDA farm statistics to calibrate these values. Details of these calculations are provided in section 11.6 of the Appendix. The uncertainty parameter κ is a scalar that multiplies the variance described in Section 2. This parameter ranges from 0.5 to 3; 0.5 represents a farmer with a relative high degree of certainty, whereas any number larger than one introduces more uncertainty to the farmer's beliefs.

[2]The stochastic nature of the function, combined with its nonlinearity, cause the output to be quite sensitive to step tolerances and optimality tolerances. Robustness checks for these parameters are detailed in Sections 11.2-11.5 of the Appendix

Estimating the coefficient of relative risk aversion in farmers has been a frequent topic of inquiry among agricultural economists, and the findings have identified a large amount of heterogeneity. Specifically, Chavas and Holt (1996) found that both corn and soybean farmers are significantly risk averse and exhibit decreasing absolute risk aversion. Moreover, Lence (2000) found that farmers' relative risk aversion has decreased throughout the 20th century; in particular, confidence intervals of risk aversion span from 1.08 to 3.67 in his estimation. Additionally, Friend and Blume (1975) estimated coefficients of relative risk aversion fall between 1 and 3. In light of these findings, we set our coefficients of relative risk aversion to fall between 1 and 3.

Table 1: Parameter Values Used in Optimization Model

Parameter	Values Considered
Risk Aversion (ρ)	$\{1,\ 2,\ 3\}$
Initial wealth (W_0)	$\{25,\ 50\}$
Optimism (o)	$\{-0.01,\ 0,\ 0.01\}$
Uncertainty (κ)	$\{0.5,\ 1,\ 1.5,\ 2,\ 3\}$
Insurance Coverage (I)	$\{0,\ 0.5,\ 0.55,\ 0.6,\ 0.65,\ 0.7,\ 0.75,\ 0.8,\ 0.85,\ 0.9\}$

6 Results

Table 2 illustrates the results of the average final hedge ratios with respect to our structural parameters for a farmer without revenue insurance. We hold each parameter constant and allow the other parameters to fluctuate while assessing the mean of the season-ending hedge ratio across 1000 iterations. Farmers who forecast a mere 1 cent increase in the futures price each week hedge substantially less than the unbiased farmer. This shows how implementing optimism into our expected utility model can rationalize a farmer's decision not to hedge their corn in the futures market. Moreover, the level of hedging increases monotonically with risk aversion. This helps explain the behavior of the small number of farmers who do hedge aggressively. Pessimistic farmers hedge the entirety of their crop across our entire grid of parameters. Unbiased farmers without insurance coverage and with more uncertainty hedge

more, as predicted by expected utility theory. Unbiased farmers with an uncertainty index of 0.5 hedge 84.38% of their crop, whereas farmers with an uncertainty level of 3 hedge nearly 10% more at 93.61%. Initial wealth effects shows that the effects of DARA hold; indeed, the optimal hedge ratio drops 3.04% as initial wealth increases from 25 to 50. Thus, the DARA results are not particularly significant at an economic level. All of these results are in accordance with our comparative statics in section 4.

Table 2: Season-Ending Hedge Ratios Across Model Specifications (Uninsured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8442	0.0064	0.9999
	2	0.8901	0.0109	1.0000
	3	0.9364	0.0571	1.0000
Uncertainty (κ)	0.5	0.8438	0.0151	1.0000
	1	0.8673	0.0116	1.0000
	1.5	0.8925	0.0110	1.0000
	2	0.9114	0.0120	1.0000
	3	0.9361	0.0742	1.0000
Wealth (W_0)	25	0.9054	0.0221	1.0000
	50	0.8750	0.0275	1.0000

Table 3 illustrates the season ending hedge ratios of our insured farmers. Across risk aversion (ρ), unbiased farmers hedge on average about 21 fewer percentage points when they have an insurance contract. This makes sense, as insurance serves as a hedge against risk, so farmers have less of an incentive to sell their corn in the futures market when covered by insurance. Insured farmers hedge less in the face of uncertainty, as explained by (16) in the comparative statics section. DARA effects still hold for insured farmers; however, similar to the uninsured farmers, the DARA effect is far from pronounced, as doubling the initial wealth only reduces hedging by 3.59 percentage points. Lastly, we see that insurance can only significantly disincentivize hedging under high coverage levels. Indeed, season-ending hedge ratios fall only 8.73 percentage points from a 50% coverage level to a 75% coverage level, and only the 90% coverage option is sufficient to significantly disincentivize

hedging. Optimistic farmers still hedge little to none of their corn in the futures market because the censored normal distribution that results from buying revenue insurance only affects the left tail. For pessimistic farmers, much of their pessimism is absorbed into the censored left tail, giving them an incentive to hedge less of their corn compared to uninsured pessimistic farmers who hedge the entirety of their crop across our grid of parameters. A seemingly counterintuitive result arises for our optimistic insured farmer when κ increases. Specifically, hedging increases with uncertainty. This is because the optimistic parameter forces the distributions away from the censored left tail of the normal distribution, thus reintroducing price risk to the farmer. As a result, optimistic farmers hedge slightly more when facing higher levels of uncertainty.

Table 3: Season-Ending Hedge Ratios Across Model Specifications (Insured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.6310	0.0046	0.8832
	2	0.6761	0.0066	0.8876
	3	0.7222	0.0391	0.8935
Uncertainty (κ)	0.5	0.7444	0.0147	1.0000
	1	0.7323	0.0104	0.9994
	1.5	0.6774	0.0086	0.8970
	2	0.6562	0.0077	0.7961
	3	0.5720	0.0424	0.7479
Wealth (W_0)	25	0.6944	0.0135	0.8905
	50	0.6585	0.0200	0.8856
Insurance (I)	0.50	0.8874	0.0244	1.0000
	0.55	0.8855	0.0241	0.9999
	0.60	0.8816	0.0237	0.9999
	0.65	0.8736	0.0231	0.9999
	0.70	0.8548	0.0216	0.9999
	0.75	0.8001	0.0154	0.9998
	0.80	0.6088	0.0083	0.9457
	0.85	0.2854	0.0065	0.6327
	0.90	0.0108	0.0040	0.4149

Figures 3-4 show the effects of different parameter combinations as they relate to the final, optimal hedge ratio. Figure 3 shows that hedging increases monotonically with both

risk aversion and uncertainty for risk averse farmers, and that the effects are synergistic. Figure 4 illustrates the magnitude of the Decreasing Absolute Risk Aversion (DARA) effect, a well known characteristic of the CRRA utility function.

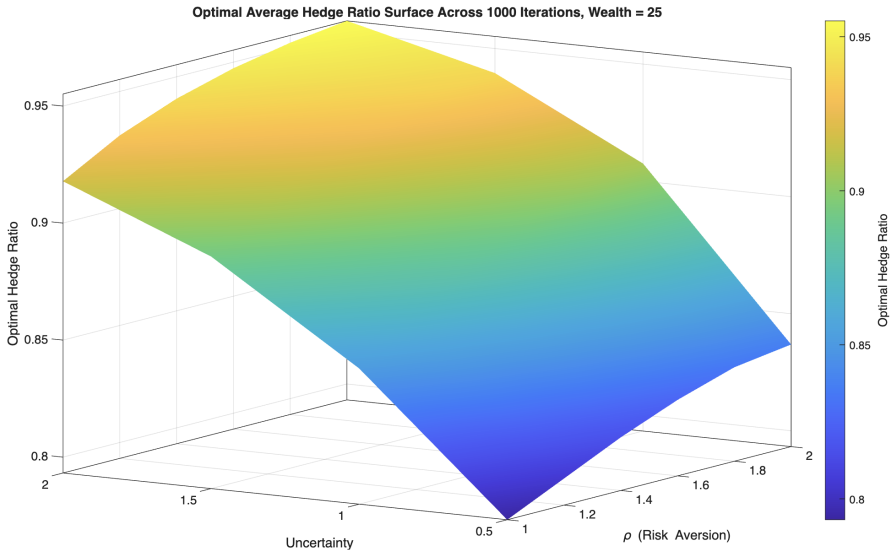


Figure 3: Risk Averse Farmers vs Uncertainty

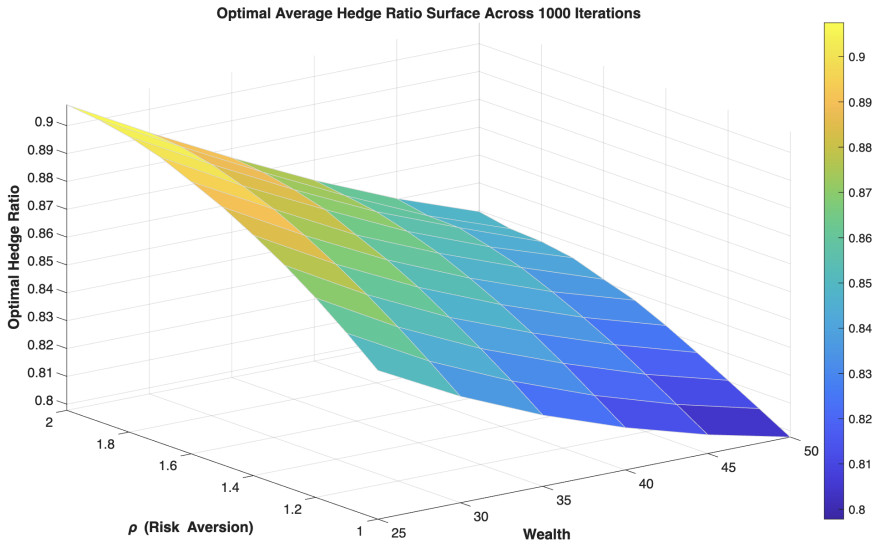
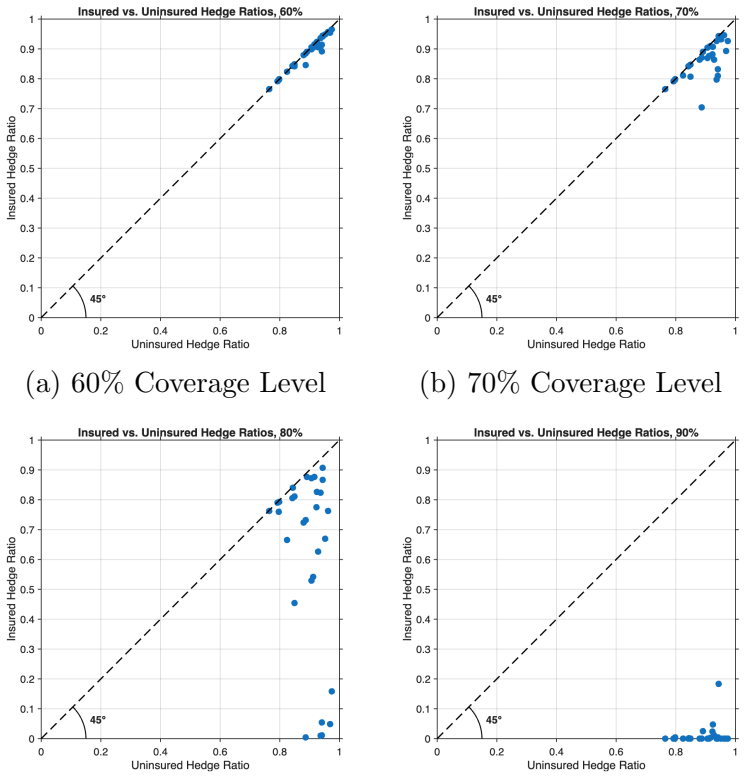


Figure 4: DARA Illustrated

7 Insurance Effects

We next analyze the crowding out effect of insurance contracts. Figure 5 illustrates the effects across 4 different insurance coverage levels: 60%, 70%, 80% and 90%. Each graph has the uninsured hedge ratio on the x-axis and the insured hedge ratio on the y-axis. Each graph has 30 different hedge ratios, representing the iterations across our 3 risk aversion levels, 5 uncertainty levels and 2 wealth levels. Figure 5 clearly illustrates the crowding out effect of insurance. In Figure 5a, most hedge ratios hover around the 45 degree line, indicating that insured and uninsured hedge ratios are close at a 60% coverage level. However, as the insurance coverage increases, we see that the optimal hedge ratios begin to favor the lower right section of the graph, indicating that the insured hedge ratio is falling. The most extreme example is illustrated in Figure 5d, where the insured hedge ratio drops to near 0, and the uninsured hedge ratio remains high.



(a) 60% Coverage Level

(b) 70% Coverage Level

(c) 80% Coverage Level

(d) 90% Coverage Level

Figure 5: Cumulative hedging under alternative insurance contracts

8 Sequential Hedging - Uninsured Farmers

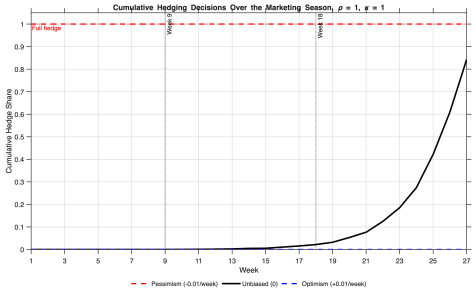
We next characterize the time path of our uninsured representative farmer's hedging decisions across different levels of risk aversion, uncertainty, and optimism/pessimsm. We analyze how these utility function parameters influence when farmers decide to sell their corn in the futures market before the December contract expires. Over 1000 iterations, we assess how their season ending hedge fluctuates across different parameters.

Under log utility ($\rho = 1$), our representative farmer hedges the majority of their corn in the futures market, but tends to wait until the end of the season to do so. The log utility is not very concave compared to the functions we will look at below. Thus, Figures 6a and 6b show that the farmer with log utility tends to forgo hedging until later in the season. We see that increasing uncertainty causes the farmer to hedge more earlier in the season. Specifically, in the first two thirds of the season, the farmer only hedges around 3% of their crop when uncertainty equals 1; however, when uncertainty is set to 3 under the same utility function, that value increases to approximately 21%. This is consistent with expected utility behavior; indeed, higher uncertainty means more potential losses, and for the risk averse farmer, the magnitude of potential losses exceeds the magnitude of potential gains.

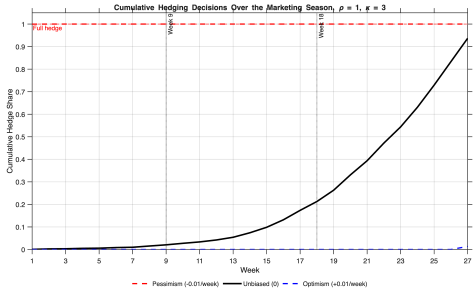
As we increase risk aversion to $\rho = 2$, and thus the concavity of the utility function, the farmer hedges earlier in the season. This is consistent with expected utility theory - early hedging locks in prices and helps alleviate uncertainty that is disliked by the risk averse farmer. When uncertainty equals 1, Figure 6c illustrates that the farmer hedges over 10% of their corn in the first 2/3 of the season. Under log utility, the farmer only hedged about 3% of their corn in the same time period. Moreover, increasing uncertainty leads to earlier hedging when $\rho = 2$, as shown in Figure 6d. Specifically, the farmer sells nearly a quarter of their crop in the futures market in the first third of the season to hedge against potential downside exposure. That number increses to 60% in the second third of the season.

Lastly, we analyze our most risk averse farmer. When $\rho = 3$, the total hedge at the end of the season is nearly identical to the season-ending hedge for a farmer with $\rho = 2$. For both

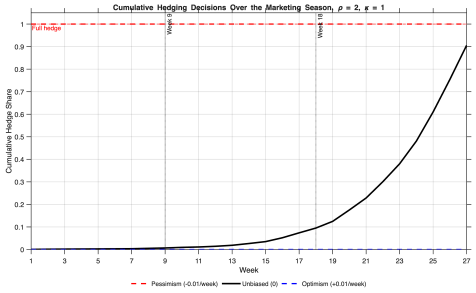
levels of risk aversion, the total hedging at the end of the season is approximately $91-96\%$, and the hedge increases to about 98% when uncertainty is increased. However, the time path of the hedge differs quite significantly. Specifically, the farmer hedges approximately 30% of their crop in the first two-thirds of the season when $\rho = 3$ and uncertainty $= 1$. When $\rho = 2$ and uncertainty $= 1$, they only hedge 11% in the same time frame.



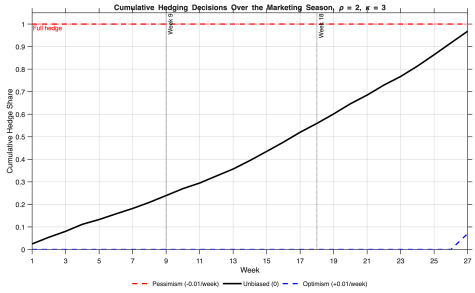
(a) $\rho = 1$, Uncertainty $= 1$



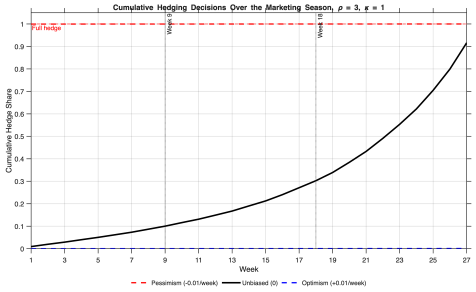
(b) $\rho = 1$, Uncertainty $= 3$



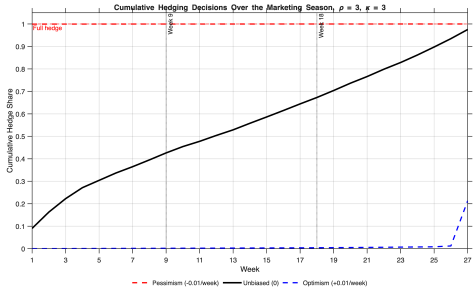
(c) $\rho = 2$, Uncertainty $= 1$



(d) $\rho = 2$, Uncertainty $= 3$



(e) $\rho = 3$, Uncertainty $= 1$



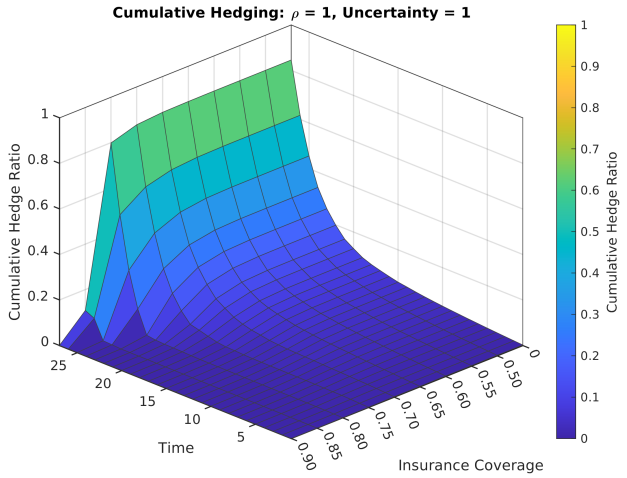
(f) $\rho = 3$, Uncertainty $= 3$

Figure 6: Cumulative hedging under alternative levels of risk aversion and uncertainty.

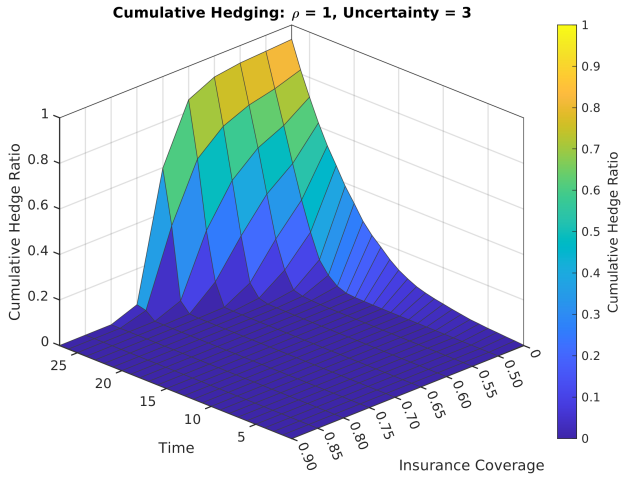
9 Sequential Hedging - Insured Farmers

We now analyze how sequential hedging decisions are impacted when our representative farmer is endowed with an insurance contract of varying levels. Per the comparative static derivations, we should expect our representative farmer to hedge less in the face of uncertainty, unlike the uninsured results. For $\rho = 1$, we see in Figure 7 that insurance delays hedging as our uncertainty parameter κ increases from 1 to 3. This result becomes much more pronounced as the insurance levels approach the upper echelon of the coverage grid; indeed, as we increase the insurance level, the mean increase becomes more amplified, thus disincentivizing hedging.

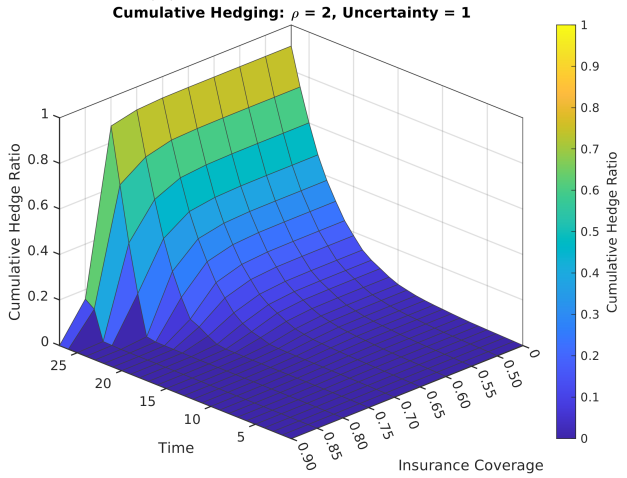
For both $\rho = 2$ and $\rho = 3$, the effects of crop insurance on hedging decisions becomes pronounced even at our smallest value of 0.50 under high levels of uncertainty. For this chosen coverage level, the season ending hedge ratio is almost identical to the uninsured farmer; however, the farmer with 0.50 coverage waits until week 18 to begin hedging. Historically, corn futures prices have already dropped at this point in the season. Thus, while both the insured and uninsured farmers hedge nearly identical amounts at small insurance coverage levels, the farmers with insurance wait until suboptimal times to sell their crop.



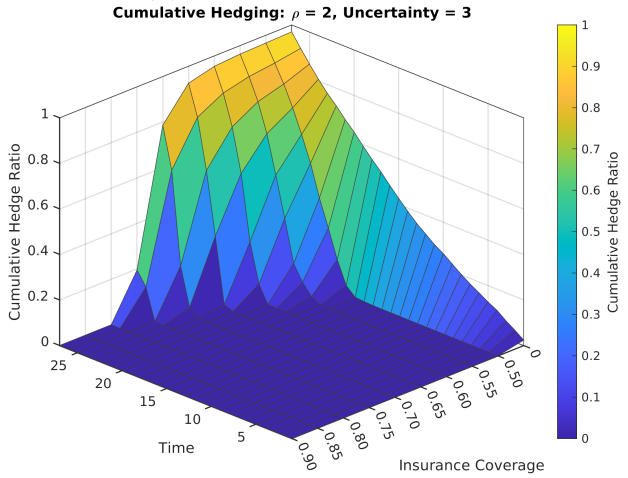
(a) $\rho = 1$, Uncertainty $= 1$



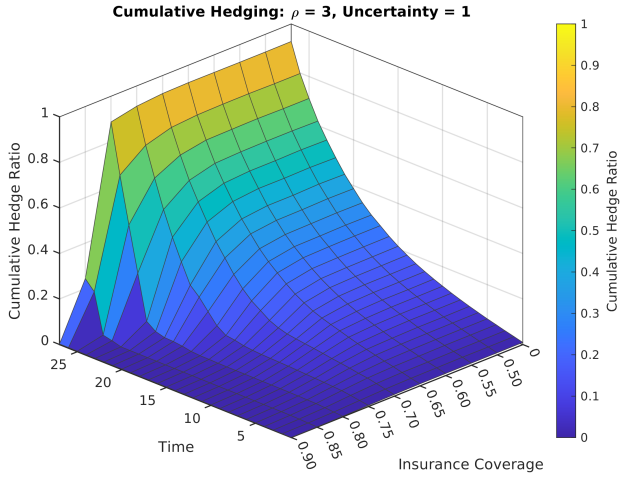
(b) $\rho = 1$, Uncertainty $= 3$



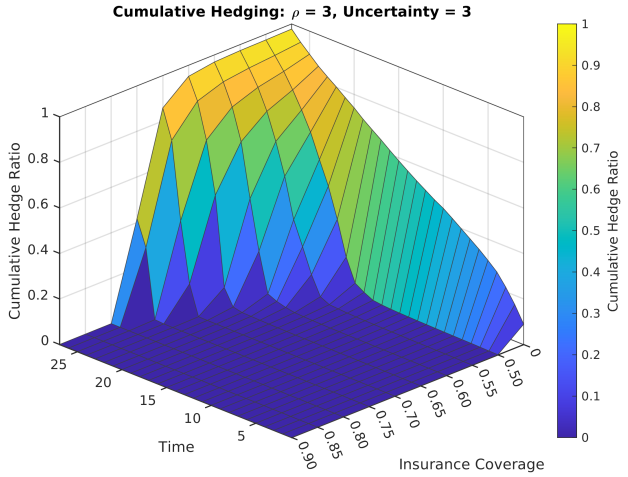
(c) $\rho = 2$, Uncertainty $= 1$



(d) $\rho = 2$, Uncertainty $= 3$



(e) $\rho = 3$, Uncertainty $= 1$

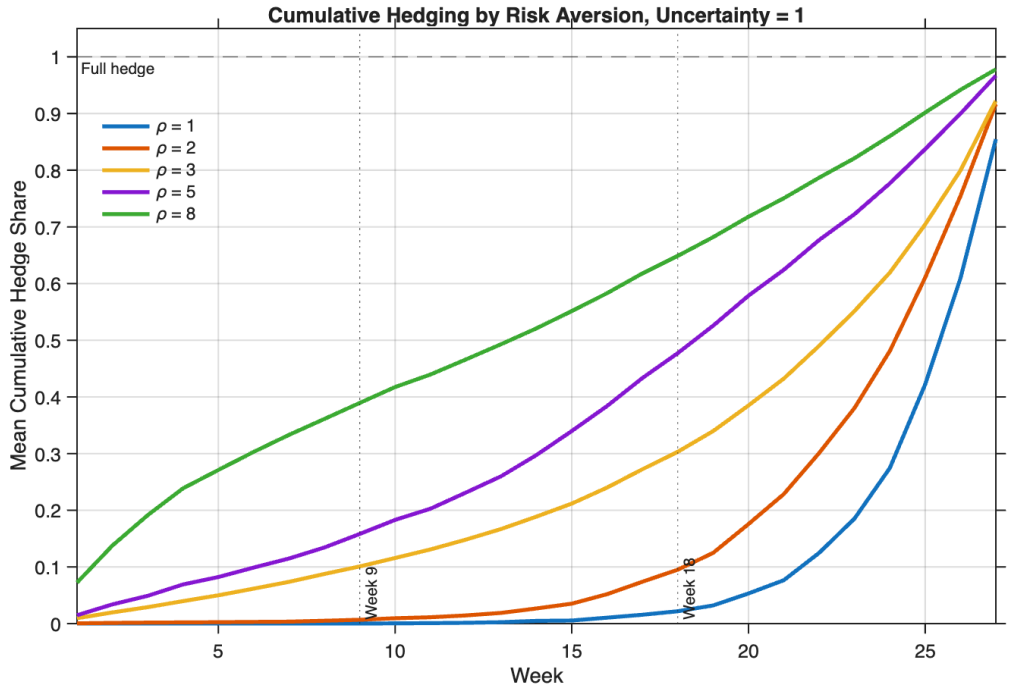


(f) $\rho = 3$, Uncertainty $= 3$

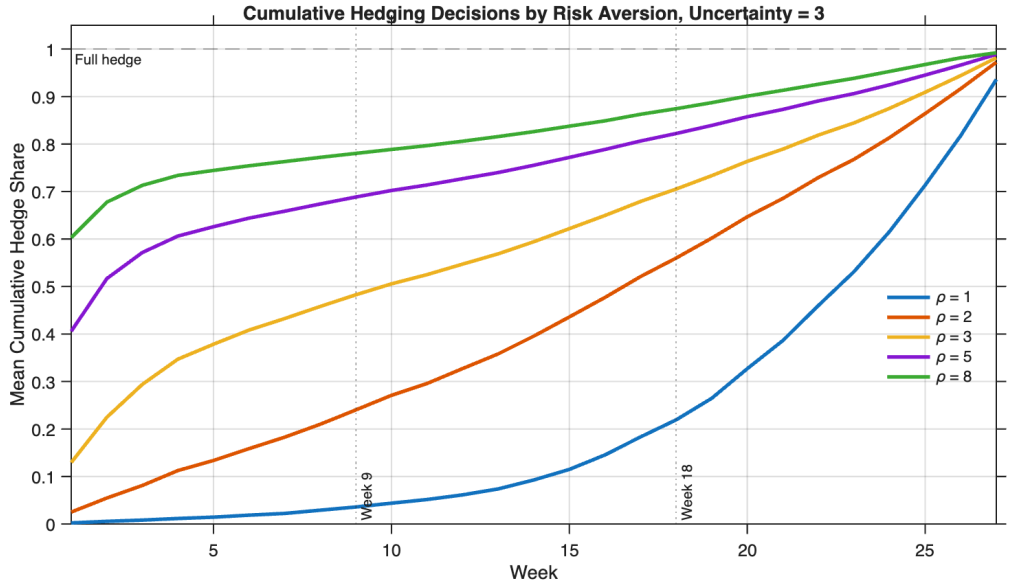
Figure 7: Cumulative hedging with insurance contracts by risk aversion and uncertainty.

10 Sensitivity Analysis

We next turn to a sensitivity analysis to see what parameter combinations are necessary to induce an unbiased farmer to hedge early in the season. Previously, we calibrated our model's risk aversion parameters to be consistent with microeconomic literature. Specifically, the studies cited above show us that farmers tend to have CRRA coefficients of approximately 1-3. In this section, we abstract away from empirical reality and assess how high risk aversion and uncertainty must be to cause significant early hedging, akin to our pessimistic farmer. Figures 8a and 8b show cumulative hedge ratios for ρ values of 1,2,3,5 and 8. Figure 8a shows that risk aversion alone is insufficient to induce early hedging, even under absurdly high risk aversion parameters. Figure 8b shows that high risk aversion, together with high uncertainty, prompt our representative farmer to hedge early on in the season; specifically, under $\rho = 5$ and $\kappa = 3$, our farmer hedges over 40% in the first week, and under $\rho = 8$ and $\kappa = 3$, the farmer hedges over 60% in the same time period. These results indicate that only an unrealistically high level of risk aversion and uncertainty can cause farmers to hedge at levels anywhere near our pessimistic farmer. The results underscore the notion that unbiased farmers, under realistic model parameters, will not hedge most of their corn until very late in the season.



(a) Uncertainty = 1



(b) Uncertainty = 3

Figure 8: Cumulative hedging by risk aversion under different levels of uncertainty.

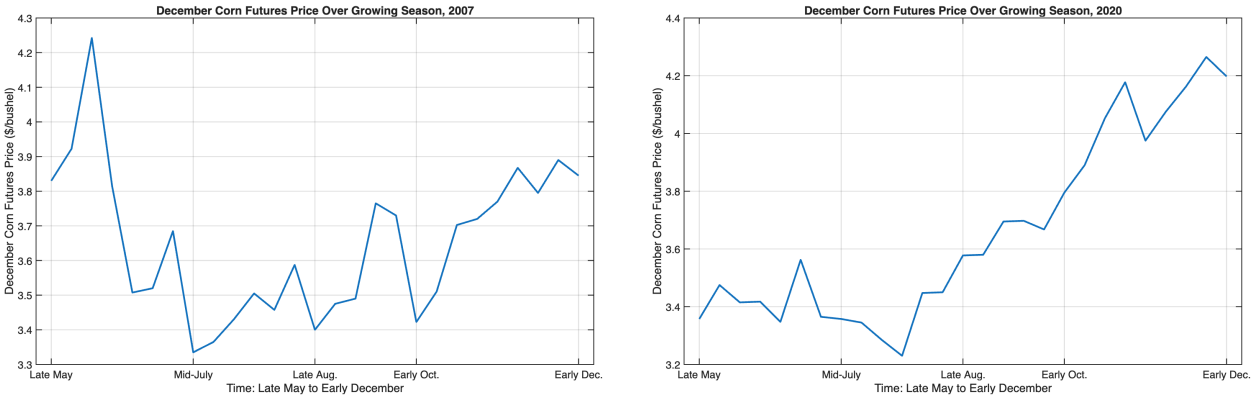
11 Robustness Check

We next analyze the same model for two additional years. In our 21 years of the December corn futures data, the futures price decreased from the first week of June to the last week of November for 16 years, and only 5 of those years showed an increase in the corn futures price in the same time period, underscoring the seasonal nature of the futures market for corn. While 2024 followed a typical trajectory of decreasing prices, we look to analyze how our representative farmer would hedge in years that display atypical patterns. For the sake of robustness, we analyze the year of 2007, where the corn price decreased, but only by 1.5 cents per bushel, representing a more muted year. We also analyze the optimal hedge ratio for 2020, where the corn futures price increased by 84 cents per bushel from the beginning of June to the end of November.

In 2007, the crop insurance guarantee was \$4.06, and the volatility factor was 0.26. We see that the uninsured hedging results are economically indistinguishable compared to our 2024 baseline analysis. However, the insured farmer in 2007 hedges more than the insured farmer in 2024, and this effect becomes especially pronounced at higher levels of insurance rates. For instance, at a 90% level insurance contract, the representative farmer hedged 13.56% in 2007; however, the farmer only hedged 1.08% in 2024 under the same contract. This reflects the fact that the unbiased farmer follows market trends, and the corn futures market in 2007 wasn't as bearish as the corn futures market in 2024, as illustrated by Figure 9a.

In 2020, the crop insurance guarantee was \$3.88, and the volatility factor was 0.15. Given the increasing nature of the corn futures price in 2020, it comes as no surprise that the unbiased insured representative farmer was less likely to hedge. Specifically, the terminal hedge ratio decreased only about 6% across the entirety of the insurance grid. In 2024, the hedge ratio fell nearly 87% across the same grid. Clearly, the corn futures prices were unusually bullish in 2020, so the unbiased insured farmer has much less of an incentive to hedge when they observe prices rising.

These results illustrate that the comparative statics hold across different years for our model parameters for uninsured farmers, and highlights the importance of analyzing hedging decisions in a sequential manner. For insured farmers, while they still hedge less when endowed with insurance, the magnitude of the insurance effect is heavily contingent upon the trend of the corn futures price. The insured farmer tends to forego hedging during years when the corn futures price is trending downward or when it is muted. However, when the corn futures price is trending upward, the unbiased farmer tends to hedge more aggressively as the price deviates upward from the crop insurance guarantee.



(a) 2007 (b) 2020

Figure 9: December Corn Futures Prices Over the Growing Season

Note: Data from Barchart

Table 4: 2007 Season-Ending Hedge Ratios Across Model Specifications (Uninsured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8582	0.0154	0.9999
	2	0.9052	0.0380	1.0000
	3	0.9413	0.1003	1.0000
Uncertainty (κ)	0.5	0.8480	0.0139	1.0000
	1	0.8799	0.0106	1.0000
	1.5	0.9074	0.0111	1.0000
	2	0.9253	0.0256	1.0000
	3	0.9471	0.1950	1.0000
Wealth (W_0)	25	0.9183	0.0608	1.0000
	50	0.8848	0.0417	1.0000

Table 5: 2007 Season-Ending Hedge Ratios Across Model Specifications (Insured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.6450	0.0089	0.8852
	2	0.7009	0.0210	0.8916
	3	0.7412	0.0613	0.8971
Uncertainty (κ)	0.5	0.8173	0.0135	1.0000
	1	0.7529	0.0093	0.9999
	1.5	0.7111	0.0076	0.8896
	2	0.6539	0.0156	0.8541
	3	0.5433	0.1059	0.7132
Wealth (W_0)	25	0.7208	0.0353	0.8945
	50	0.6707	0.0255	0.8881
Insurance	0.50	0.8947	0.0503	0.9999
	0.55	0.8906	0.0499	0.9999
	0.60	0.8831	0.0488	0.9999
	0.65	0.8702	0.0464	0.9999
	0.70	0.8433	0.0391	0.9998
	0.75	0.7679	0.0217	0.9970
	0.80	0.6075	0.0078	0.8862
	0.85	0.3685	0.0056	0.7373
	0.90	0.1356	0.0040	0.4021

Table 6: 2020 Season-Ending Hedge Ratios Across Model Specifications (Uninsured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8312	0.0040	0.9999
	2	0.8737	0.0023	1.0000
	3	0.9315	0.0351	1.0000
Uncertainty (κ)	0.5	0.8453	0.0163	1.0000
	1	0.8571	0.0131	1.0000
	1.5	0.8759	0.0109	1.0000
	2	0.8952	0.0102	1.0000
	3	0.9204	0.0185	1.0000
Wealth (W_0)	25	0.8907	0.0041	1.0000
	50	0.8669	0.0235	1.0000

Table 7: 2020 Season-Ending Hedge Ratios Across Model Specifications (Insured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8125	0.0038	0.9999
	2	0.8618	0.0021	1.0000
	3	0.9115	0.0328	0.9999
Uncertainty (κ)	0.5	0.8375	0.0163	1.0000
	1	0.8486	0.0130	0.9999
	1.5	0.8645	0.0106	0.9999
	2	0.8775	0.0095	0.9999
	3	0.8814	0.0151	0.9999
Wealth (W_0)	25	0.8794	0.0038	1.0000
	50	0.8444	0.0220	0.9999
Insurance	0.50	0.8780	0.0137	1.0000
	0.55	0.8776	0.0136	1.0000
	0.60	0.8766	0.0135	1.0000
	0.65	0.8750	0.0133	0.9999
	0.70	0.8722	0.0131	0.9999
	0.75	0.8671	0.0129	0.9999
	0.80	0.8576	0.0125	0.9999
	0.85	0.8418	0.0122	0.9999
	0.90	0.8113	0.0114	0.9999

12 Conclusion

In the hedging literature, much attention has been paid to the terminal hedge ratio at the end of the season; however, little attention has been paid to the optimal timing of hedging decisions, given the seasonal nature of futures prices. Moreover, it is well documented in the literature that farmers do not hedge as much as economists recommended. Our results show that an unbiased representative farmer forgoes hedging until late in the harvesting season, even under high levels of risk aversion and uncertainty. Endowing our farmer with a tepid amount of optimism is sufficient to forgo hedging until much later in the season. Only a pessimistic farmer who predicts a 1 cent decrease in the weekly futures price hedges the entirety of their crop in week 1. Insured farmers also delay hedging, and farmers with a high

level of insurance forgo hedging completely, underscoring the substitutability of insurance contracts and hedging. The effect of uncertainty on hedging decisions is contingent upon insurance contracts. Uninsured farmers hedge more in the face of uncertainty; specifically, terminal hedge ratios increase by approximately 9% when we increase uncertainty from 0.5 to 3. Additionally, insured farmers hedge less, due to the mean amplifying effects of variance multiplication under left-censored prices. Indeed, optimal hedge ratios decrease by about 17% across our grid of uncertainty parameters. Risk averse farmers hedge about 9% more from $\rho = 1$ to $\rho = 3$ across both insured and uninsured farmers. While the DARA effects of the CRRA utility function are present, the effects are neglibile, as doubling wealth only decreases the optimal hedge ratio by about 3% across both insured and uninsured farmers. A sensitivity analysis reveals that only a farmer with an absurd level of risk aversion and uncertainty would hedge the majority of their crop in the first week of our time period. Thus, building a more complete and realistic model of farmer marketing and risk management decisions solves the mystery of why farmers hedge so much less than agricultural econoimsts recommend.

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13 Appendix

13.1 Proof of Equation 13

Let $z = \sigma^2$, so (12) can be expressed in terms of variance instead of standard deviation, namely, $\sigma = \sqrt{z}$. Let $\alpha = \frac{a-\mu}{\sqrt{z}}$, $\Phi = \Phi(\alpha)$, $\phi = \frac{1}{\sqrt{2\pi}} \exp \frac{1}{2}\alpha^2$. Lastly, let λ be the inverse Mills ratio, $\frac{\phi}{1-\Phi}$. Recall $\mathbb{E}[y] = \Phi a + (1 - \Phi)(\mu + \sigma\lambda)$.

$$\frac{\partial \mathbb{E}[y]}{\partial \sigma^2} = \frac{\partial \Phi}{\partial \sigma^2} a + (1-\Phi)\sqrt{\sigma^2}\frac{\partial \lambda}{\partial z} + \underbrace{\frac{(1-\Phi)\lambda}{2\sqrt{\sigma^2}}}_{=\frac{\phi}{2\sqrt{\sigma^2}}} - \frac{\partial \Phi}{\partial \sigma^2}(\mu + \sqrt{\sigma^2}\lambda)$$

Because the Inverse Mills Ratio has σ^2 in both the numerator and the denominator, we appeal to the quotient rule:

$$\frac{\partial \lambda}{\partial \sigma^2} = \frac{(1-\Phi)\frac{\partial \phi}{\partial \sigma^2} - \phi\frac{\partial (1-\Phi)}{\partial \sigma^2}}{(1-\Phi)^2}$$

Plugging this result in yields:

$$= \frac{\partial \Phi}{\partial \sigma^2} a + (1-\Phi)\sqrt{\sigma^2}\left(\frac{(1-\Phi)\frac{\partial \phi}{\partial \sigma^2} - \phi\frac{\partial (1-\Phi)}{\partial \sigma^2}}{(1-\Phi)^2}\right) + \frac{\phi}{2\sqrt{\sigma^2}} - \frac{\partial \Phi}{\partial \sigma^2}(\mu + \sqrt{\sigma^2}\lambda)$$

We distribute $(1-\Phi)$, and exploit the fact that the Inverse Mills Ratio is found in our derivation of the quotient rule:

$$= \frac{\partial \Phi}{\partial \sigma^2} a + \frac{\sqrt{\sigma^2}\frac{\partial \phi}{\partial \sigma^2}(1-\Phi)^2}{(1-\Phi)^2} + \underbrace{\frac{\sqrt{\sigma^2}\phi\frac{\partial \Phi}{\partial \sigma^2}}{(1-\Phi)}}_{=\sqrt{\sigma^2}\lambda\frac{\partial \Phi}{\partial \sigma^2}} - \frac{\partial \Phi}{\partial \sigma^2}\mu - \sqrt{\sigma^2}\lambda\frac{\partial \Phi}{\partial \sigma^2} + \frac{\phi}{2\sqrt{\sigma^2}}$$

$\sqrt{\sigma^2}\lambda\frac{\partial \Phi}{\partial \sigma^2}$ and -$\sqrt{\sigma^2}\lambda\frac{\partial \Phi}{\partial \sigma^2}$ cancel out.

$$= \frac{\partial \Phi}{\partial \sigma^2} a + \sqrt{\sigma^2}\frac{\partial \phi}{\partial \sigma^2} - \frac{\partial \Phi}{\partial \sigma^2}\mu + \frac{\phi}{2\sqrt{\sigma^2}}$$

Recall $\Phi = \Phi(\frac{a-\mu}{\sqrt{z}})$ and $\phi = \frac{1}{\sqrt{2\pi}}\exp(-\frac{1}{2}(\frac{a-\mu}{\sqrt{z}})^2)$. We next differentiate these two functions with respect to σ^2

$$\frac{\partial \Phi}{\partial \sigma^2} = -\frac{a-\mu}{2(\sigma^2)^{\frac{3}{2}}}\phi = \frac{a-\mu}{2 * \sqrt{\sigma^2} * \sigma^2}\phi = -\frac{\alpha}{2\sigma^2}\phi$$

$$\frac{\partial \phi}{\partial \sigma^2} = \frac{(a-\mu)^2}{2(\sigma^2)^2}\phi = \frac{(a-\mu)^2}{2 * \sqrt{\sigma^2} * (\sigma^2)^{1.5}} = \frac{\alpha(a-\mu)}{2(\sigma^2)^{1.5}}$$

Plugging in yields:

$$= \frac{-\alpha}{2\sigma^2}\phi a + \sqrt{\sigma^2}\frac{\alpha(a-\mu)}{2(\sigma^2)^{1.5}} + \frac{\alpha}{2\sigma^2}\phi\mu + \frac{\phi}{2\sqrt{\sigma^2}}$$

This algebraically simplifes to:

$$= \frac{\alpha}{2\sqrt{\sigma^2}}\phi(\frac{a-\mu}{\sqrt{\sigma^2}}) - \frac{\alpha}{2\sqrt{\sigma^2}}\phi(\frac{a-\mu}{\sqrt{\sigma^2}}) + \frac{\phi}{2\sqrt{\sigma^2}}$$

The first two terms cancel out, leaving us with a surprisingly elegant result.

$$= \frac{\phi}{2\sqrt{\sigma^2}} = \frac{\phi}{2\sigma} \quad \blacksquare$$

13.2 Gradient Function

13.2.1 Gradient of the Objective Function

MATLAB's fmincon package is flexible towards specifying an objective gradient function. This improves both the efficiency of the algorithm and the accuracy of the optimization. The gradient function for each time period t is defined as such:

Revenue in period t is

$$R = W_0 + \sum_{t=1}^{T} f_t\theta_t + \sum_{k=1}^{t-1} f_k\theta_k + \sum_{j=t+1}^{T-1} (\tilde{f}_j + o)\tilde{\theta}_j + \left(1 - \sum_{t=1}^{T}\theta_t - \sum_{j=t+1}^{T-1}\tilde{\theta}_j\right)\tilde{P}_T.$$

Expected utility is

$$f(\theta, \tilde{\theta}) = \mathbb{E}\left[\frac{R^{1-\rho}}{1-\rho}\right].$$

The gradient used by the optimizer is

$$\frac{\partial f}{\partial \theta_t} = \mathbb{E}\Big[(f_t - \tilde{P}_T)R^{-\rho}\Big]$$

and for forward sales $j = t+1, \ldots, T-1$,

$$\frac{\partial f}{\partial \tilde{\theta}_j} = \mathbb{E}\Big[(\tilde{f}_j + o - \tilde{P}_T)R^{-\rho}\Big].$$

13.3 Optimization Parameters in MATLAB

The representative farmer's utility function becomes very flat at high levels of risk aversion. Thus, the default optimization parameters in MATLAB fail to provide meaningful results. We provide a sensitivity analysis of these parameters to find optimal values.

13.4 Optimality Tolerance

Let $\varepsilon_{\text{OptTol}}$ denote the optimality tolerance, $\mathscr{L}$ be the Lagrangian, and θ be the hedge ratio chosen by the farmer. If

$$\max_i \|\frac{\partial \mathscr{L}}{\partial \theta_i}\| \leq \epsilon_{OptTol} \tag{17}$$

Then the iterations stop. MATLAB's default value for optimality tolerance equals $1e^{-6}$. The results below show that the mean hedge tends to stability around OptTol of $1e^{-10}$ across all risk aversion parameters. Thus, we change the value of OptTol to this value.

13.4.1 $\rho = 1$

Table 8: Optimality Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 1$, $W_0 = 25$)

ρ	Optimism	OptTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
1	-0.01	1.00×10^{-6}	1.0000	–	–	Baseline
1	-0.01	1.00×10^{-12}	1.0000	0	1.05×10^{-5}	
1	-0.01	1.00×10^{-10}	1.0000	0	1.05×10^{-5}	
1	-0.01	1.00×10^{-8}	1.0000	0	1.05×10^{-5}	
1	0	1.00×10^{-6}	0.8166	–	–	Baseline
1	0	1.00×10^{-12}	0.8198	3.20×10^{-3}	2.69×10^{-4}	
1	0	1.00×10^{-10}	0.8198	3.20×10^{-3}	2.69×10^{-4}	
1	0	1.00×10^{-8}	0.8198	3.20×10^{-3}	2.66×10^{-4}	
1	0.01	1.00×10^{-6}	0.0039	–	–	Baseline
1	0.01	1.00×10^{-12}	1.77×10^{-5}	-3.88×10^{-3}	5.21×10^{-6}	
1	0.01	1.00×10^{-10}	1.72×10^{-5}	-3.88×10^{-3}	5.21×10^{-6}	
1	0.01	1.00×10^{-8}	4.88×10^{-5}	-3.85×10^{-3}	5.18×10^{-6}	

13.4.2 $\rho = 2$

Table 9: Optimality Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 2$, $W_0 = 25$)

ρ	Optimism	OptTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
2	-0.01	1.00×10^{-6}	0.9998	–	–	Baseline
2	-0.01	1.00×10^{-12}	1.0000	0.0002	1.03×10^{-5}	
2	-0.01	1.00×10^{-10}	1.0000	0.0002	NaN	
2	-0.01	1.00×10^{-8}	1.0000	0.0002	1.02×10^{-5}	
2	0	1.00×10^{-6}	0.9408	–	–	Baseline
2	0	1.00×10^{-12}	0.8943	-0.0465	1.72×10^{-5}	
2	0	1.00×10^{-10}	0.8942	-0.0466	1.70×10^{-5}	
2	0	1.00×10^{-8}	0.8856	-0.0552	1.83×10^{-5}	
2	0.01	1.00×10^{-6}	0.1815	–	–	Baseline
2	0.01	1.00×10^{-12}	4.36×10^{-6}	-0.1815	5.91×10^{-6}	
2	0.01	1.00×10^{-10}	1.27×10^{-5}	-0.1815	5.91×10^{-6}	
2	0.01	1.00×10^{-8}	0.0010	-0.1805	5.90×10^{-6}	

13.4.3 $\rho = 3$

Table 10: Optimality Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 3$, $W_0 = 25$)

ρ	Optimism	OptTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
3	-0.01	1.00×10^{-6}	0.9926	–	–	Baseline
3	-0.01	1.00×10^{-12}	1.0000	0.0074	1.41×10^{-5}	
3	-0.01	1.00×10^{-10}	1.0000	0.0074	1.41×10^{-5}	
3	-0.01	1.00×10^{-8}	0.9998	0.0072	1.36×10^{-5}	
3	0	1.00×10^{-6}	0.9724	–	–	Baseline
3	0	1.00×10^{-12}	0.9173	-0.0551	-3.97×10^{-7}	
3	0	1.00×10^{-10}	0.9181	-0.0543	-3.51×10^{-7}	
3	0	1.00×10^{-8}	0.9303	-0.0421	-2.72×10^{-7}	
3	0.01	1.00×10^{-6}	0.8700	–	–	Baseline
3	0.01	1.00×10^{-12}	5.94×10^{-4}	-0.8694	1.30×10^{-6}	
3	0.01	1.00×10^{-10}	0.0010	-0.8690	1.30×10^{-6}	
3	0.01	1.00×10^{-8}	0.0542	-0.8158	1.24×10^{-6}	

13.5 Step Tolerances

Let $\varepsilon_{\text{StepTol}}$ be the step tolerance. Step tolerance denotes the lower bound of the magnitude of a step on the x axis. Namely, $\|x_{t+1} - x_t\|_\infty$. In the fmincon package we employ, Step Tolerance is relative bound. Thus, MATLAB halts the optimization algorithm when

$$\|x_{t+1} - x_t\|_\infty < \varepsilon_{\text{StepTol}} * (1 + |x_i|) \tag{18}$$

MATLAB's default value for step tolerances = $1e^{-10}$. The algorithm, as illustrated below, does not seem to sensitive to step tolerance levels. Thus, the default value is chosen.

13.5.1 $\rho = 1$

Table 11: Step Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 1$, $W_0 = 25$)

ρ	Optimism	StepTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
1	-0.01	1.00×10^{-10}	1.0000	–	–	Baseline
1	-0.01	1.00×10^{-12}	1.0000	0	0	
1	-0.01	1.00×10^{-8}	1.0000	0	-9.95×10^{-6}	
1	-0.01	1.00×10^{-6}	0.9995	-5.00×10^{-4}	NaN	
1	0	1.00×10^{-10}	0.8166	–	–	Baseline
1	0	1.00×10^{-12}	0.8166	0	0	
1	0	1.00×10^{-8}	0.8166	0	0	
1	0	1.00×10^{-6}	0.8166	0	0	
1	0.01	1.00×10^{-10}	0.0039	–	–	Baseline
1	0.01	1.00×10^{-12}	0.0039	0	0	
1	0.01	1.00×10^{-8}	0.0039	0	0	
1	0.01	1.00×10^{-6}	0.0040	1.00×10^{-4}	-9.24×10^{-7}	

13.5.2 $\rho = 2$

Table 12: Step Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 2$, $W_0 = 25$)

ρ	Optimism	StepTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
2	-0.01	1.00×10^{-10}	0.9998	–	–	Baseline
2	-0.01	1.00×10^{-12}	0.9998	0	0	
2	-0.01	1.00×10^{-8}	0.9998	0	0	
2	-0.01	1.00×10^{-6}	0.9993	-5.00×10^{-4}	-7.62×10^{-5}	
2	0	1.00×10^{-10}	0.9408	–	–	Baseline
2	0	1.00×10^{-12}	0.9408	0	0	
2	0	1.00×10^{-8}	0.9408	0	0	
2	0	1.00×10^{-6}	0.9498	9.00×10^{-3}	-1.27×10^{-6}	
2	0.01	1.00×10^{-10}	0.1815	–	–	Baseline
2	0.01	1.00×10^{-12}	0.1815	0	0	
2	0.01	1.00×10^{-8}	0.1815	0	0	
2	0.01	1.00×10^{-6}	0.1815	0	0	

13.5.3 $\rho = 3$

Table 13: Step Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 3$, $W_0 = 25$)

ρ	Optimism	StepTol	Mean Hedge	Δ Mean Hedge	Δ Expected Utility	Notes
3	-0.01	1.00×10^{-10}	0.9926	–	–	Baseline
3	-0.01	1.00×10^{-12}	0.9926	0	0	
3	-0.01	1.00×10^{-8}	0.9926	0	0	
3	-0.01	1.00×10^{-6}	0.9895	-3.10×10^{-3}	3.30×10^{-7}	
3	0	1.00×10^{-10}	0.9724	–	–	Baseline
3	0	1.00×10^{-12}	0.9724	0	0	
3	0	1.00×10^{-8}	0.9724	0	0	
3	0	1.00×10^{-6}	0.9724	0	0	
3	0.01	1.00×10^{-10}	0.8700	–	–	Baseline
3	0.01	1.00×10^{-12}	0.8700	0	0	
3	0.01	1.00×10^{-8}	0.8700	0	0	
3	0.01	1.00×10^{-6}	0.9856	1.156×10^{-1}	-5.54×10^{-7}	

13.6 Prior Sample Size Robustness Check

13.7 5,000 Prior Draws

Table 14: Season-Ending Hedge Ratios Across Model Specifications (Uninsured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8183	0.0109	0.9999
	2	0.8657	0.0170	1.0000
	3	0.9165	0.0631	1.0000
Uncertainty (κ)	0.5	0.8186	0.0149	1.0000
	1	0.8423	0.0118	1.0000
	1.5	0.8679	0.0112	1.0000
	2	0.8884	0.0159	1.0000
	3	0.9171	0.0979	1.0000
Wealth (W_0)	25	0.8829	0.0277	1.0000
	50	0.8508	0.0330	1.0000

Table 15: Season-Ending Hedge Ratios Across Model Specifications (Insured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.6122	0.0073	0.8821
	2	0.6554	0.0105	0.8858
	3	0.7052	0.0437	0.8910
Uncertainty (κ)	0.5	0.7233	0.0145	1.0000
	1	0.7113	0.0107	0.9994
	1.5	0.6613	0.0092	0.8939
	2	0.6364	0.0108	0.7988
	3	0.5557	0.0574	0.7395
Wealth (W_0)	25	0.6745	0.0172	0.8883
	50	0.6407	0.0238	0.8844
Insurance (I)	0.50	0.8641	0.0300	1.0000
	0.55	0.8618	0.0298	0.9999
	0.60	0.8574	0.0295	0.9999
	0.65	0.8491	0.0287	0.9999
	0.70	0.8281	0.0269	0.9999
	0.75	0.7716	0.0195	0.9996
	0.80	0.5918	0.0090	0.9306
	0.85	0.2837	0.0069	0.6376
	0.90	0.0109	0.0043	0.4093

13.8 50,000 Prior Draws

Table 16: Season-Ending Hedge Ratios Across Model Specifications (Uninsured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.8895	0.0039	0.9999
	2	0.9274	0.0023	1.0000
	3	0.9568	0.0403	1.0000
Uncertainty (κ)	0.5	0.8910	0.0132	1.0000
	1	0.9055	0.0092	1.0000
	1.5	0.9255	0.0089	1.0000
	2	0.9408	0.0094	1.0000
	3	0.9599	0.0368	1.0000
Wealth (W_0)	25	0.9369	0.0112	1.0000
	50	0.9122	0.0198	1.0000

Table 17: Season-Ending Hedge Ratios Across Model Specifications (Insured)

Category	Parameter	Unbiased	Optimism	Pessimism
Risk Aversion (ρ)	1	0.6495	0.0031	0.8821
	2	0.6997	0.0016	0.8876
	3	0.7291	0.0263	0.8940
Uncertainty (κ)	0.5	0.7797	0.0126	1.0000
	1	0.7478	0.0084	0.9994
	1.5	0.6916	0.0066	0.9024
	2	0.6665	0.0053	0.7844
	3	0.5782	0.0187	0.7532
Wealth (W_0)	25	0.7127	0.0067	0.8911
	50	0.6728	0.0140	0.8846
Insurance (I)	0.50	0.9202	0.0151	0.9999
	0.55	0.9175	0.0146	0.9999
	0.60	0.9131	0.0141	1.0000
	0.65	0.9045	0.0134	0.9999
	0.70	0.8849	0.0124	0.9999
	0.75	0.8177	0.0090	0.9998
	0.80	0.6049	0.0065	0.9554
	0.85	0.2629	0.0050	0.6117
	0.90	0.0092	0.0029	0.4244

13.9 Initial Wealth Calculation

We assume that farmers' initial wealth is a function of land value and investments in farming equipment. In 2024, corn farmers yielded about 179.3 bushels per acre (USDA, NASS (2025)). Additionally, the average corn farm contained about 466 acres on average (Lacy (2025)). Thus, total bushels farmed on average equates to 179.3 * 466 = 83,553.80. Moreover, land value was found to be approximately $4,170 per acre (Callahan (2026)). Machinery investment value was found to be $847 per acre for farms that were less than 500 acres (Langemeier (2025)). Thus, total land value can be appoximated as 466 * $4,170 = $1,943,220. Total equipment value can be calculated as $847 * 466 = $394,702. After combining land value and equipment value, our representative farmer has a total wealth of $1,943,220 + $394,702 = $2,337,922. Thus, wealth per bushel can be approximated as

$2,337,922 / 83,553.80 ≈ $27.98. Thus, we simplify and use $25 as a baseline wealth level. Clearly, farmers often have income from outside of the farm as well, and some farmland is much more expensive than $4,000 per acre. Thus, we use $50 as the initial wealth level for farmers with substantial off-farm assets or more expensive farmland.